Practical 14 - RAW Socket Packet

AIM

To write a program using RAW sockets to capture and display network packet information.

ALGORITHM

1. Import the socket module.
2. Create a raw socket using the appropriate address family and protocol.
3. Run the program with the required permissions.
4. Capture packets arriving at the system's network interface.
5. Display packet source information and packet length.
6. Stop capture and close the socket.

PYTHON CODE - WINDOWS

```
import socket

s = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
s.bind(("0.0.0.0", 0))
s.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)
s.ioctl(socket.SIO_RCVALL, socket.RCVALL_ON)

print("Packet sniffing started...")
try:
    while True:
        data, addr = s.recvfrom(65535)
        print("Packet received from:", addr[0])
        print("Packet length:", len(data))
        print("-" * 40)
except KeyboardInterrupt:
    print("\nSniffing stopped.")
finally:
    s.ioctl(socket.SIO_RCVALL, socket.RCVALL_OFF)
    s.close()
```

INPUT

Network packets visible to the selected system network interface.

OUTPUT

The program displays the source IP address and length of each captured packet.

NOTE

This example uses Windows-specific raw-socket behavior and generally must be run as Administrator. Capture only traffic you are authorized to inspect.

RESULT

A basic packet-sniffing program using raw sockets was implemented to capture and display packet information.